

1.1.3 Gears

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Graphs

Number of Teeth per Gear				
	A (n_{in})	B	C	D
n (teeth)	36T	60T	60T	84T

	B:A	C:B	D:C		
Gear Ratio (reduced)	5:3	1:1	7:5		
Gear Ratios as Fractions	5/3	1/1	7/5	Product of Fractions	Final Gear Ratio
				7/3	7:3

Questions

How many times will gear A rotate compared to gear D? How Does this compare to the gear ratio you just calculated in the table?

For every 1 time gear A rotates, gear D rotates 0.4 times.

If a 10 ft-lbs of torque is applied at gear A, then what is the output at gear D?

30/7 ft-lbs or ~ 4.275 ft-lbs

How can you make gear A and D rotate in the same direction?

Switch gear A with gear B or gear D with gear C.

Questions (continued)

What will the gear ratio be if A is connected to D

3:7

How does the ratio between A and D compare to that of the entire system?

Equivalent

If gear D is the final output, or where the load is attached, then how did gear B and C impact the system?

It didn't, you cross the same amount of teeth regardless of other gears in between. You just divide the total number of teeth crossed by the amount of teeth in the output gear.